# Sort a K-Sorted Array

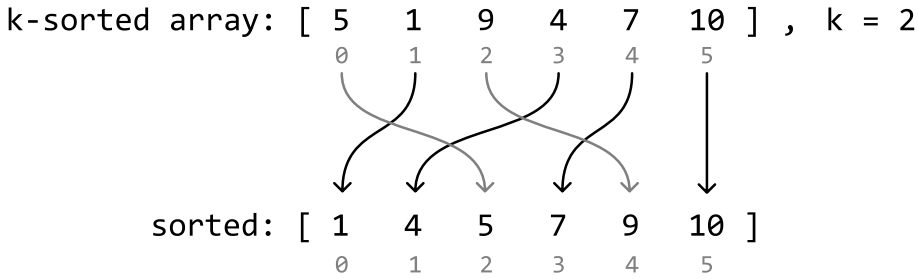
Given an integer array where each element is at most `k` positions away from its sorted position, **sort the array** in a non-decreasing order.

## Example:

```
Input: nums = [5, 1, 9, 4, 7, 10], k = 2
Output: [1, 4, 5, 7, 9, 10]
```

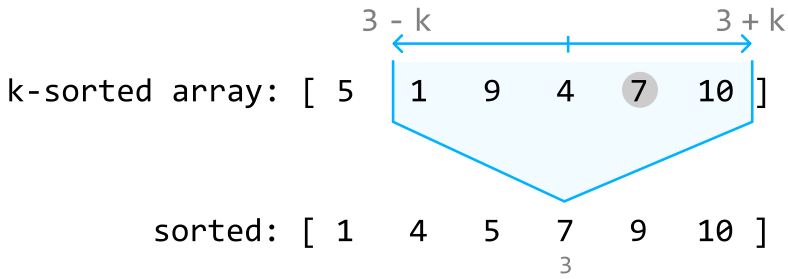
# Intuition

In a k-sorted array, each element is at most k indexes away from where it would be in a fully sorted array. We can visualize this with the following example where k = 2, and no number is more than k indexes away from its sorted position:

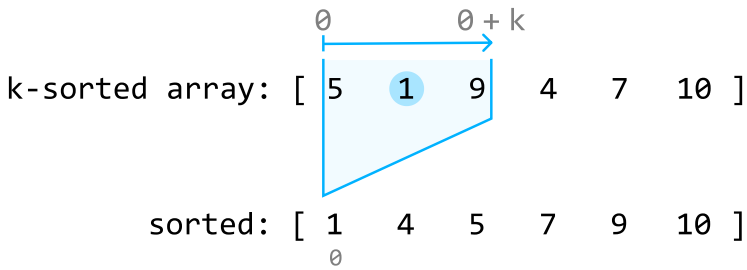



A trivial solution to this problem is to sort the array using a standard sorting algorithm. However, since the input is partially sorted (k-sorted), we should assume there's a faster way to sort the array.

We can think about this problem backward. For any index `i`, **the element that belongs at index `i` in the sorted array is located within the range [`i` - k, `i` + k]**. Below, we visualize how number 7, which is meant to be at index 3 when sorted, correctly falls within the range [3 - k, 3 + k] in the k-sorted array:

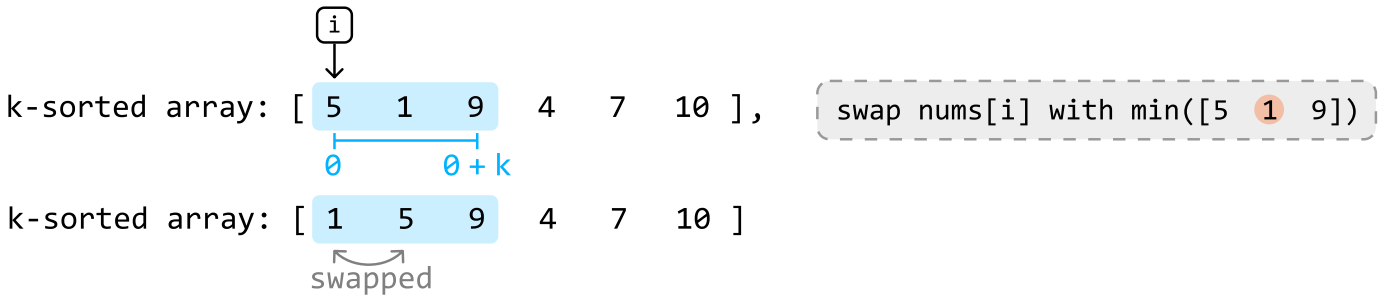



This is a good start, but we can reduce this range even further. Consider index 0 from the above array. We know the number which belongs at index 0 when sorted is somewhere in the range [0, 0 + k]:



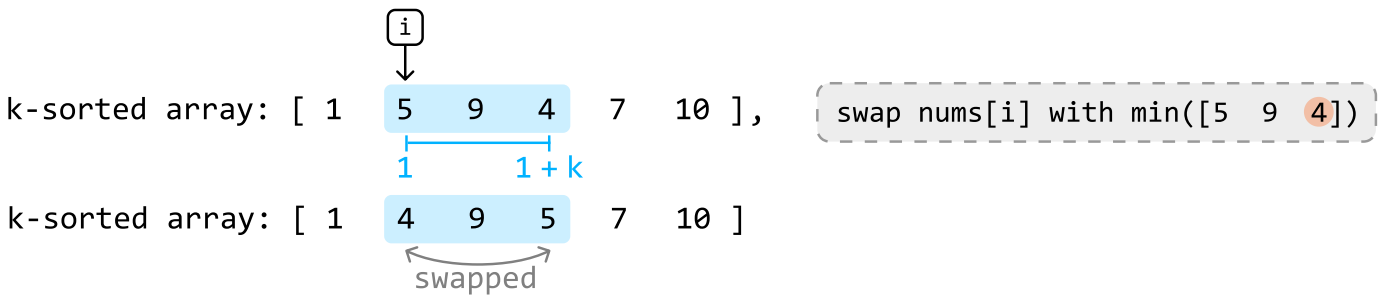

Note that the sorted array in the diagrams is purely provided as a reference point. We don't yet know which number in the range [0, 0 + k] belongs at index 0. However, one fact remains consistent: in a sorted array, index 0 always holds the smallest number. This means the value needed at index 0 is also the smallest number within the range [0, 0 + k] of the k-sorted array, which is 1 in this example.

So, let's swap 1 with the number at index 0 to position 1 as the first value in the sorted array:




Now let's find the number that belongs at index 1: the second smallest number. Since index 0 currently contains the smallest value in the array, we won't need to consider index 0 in our search. Therefore, we can find the value that belongs at index 1 in the range [1, 1 + k]. The smallest value in this range will be the second smallest value overall, which is 4 in this case.

So, let's swap 4 with the number at index 1 to position 4 as the second value in the sorted array:




If we continue this process for the rest of the array, we'll successfully sort the k-sorted array.

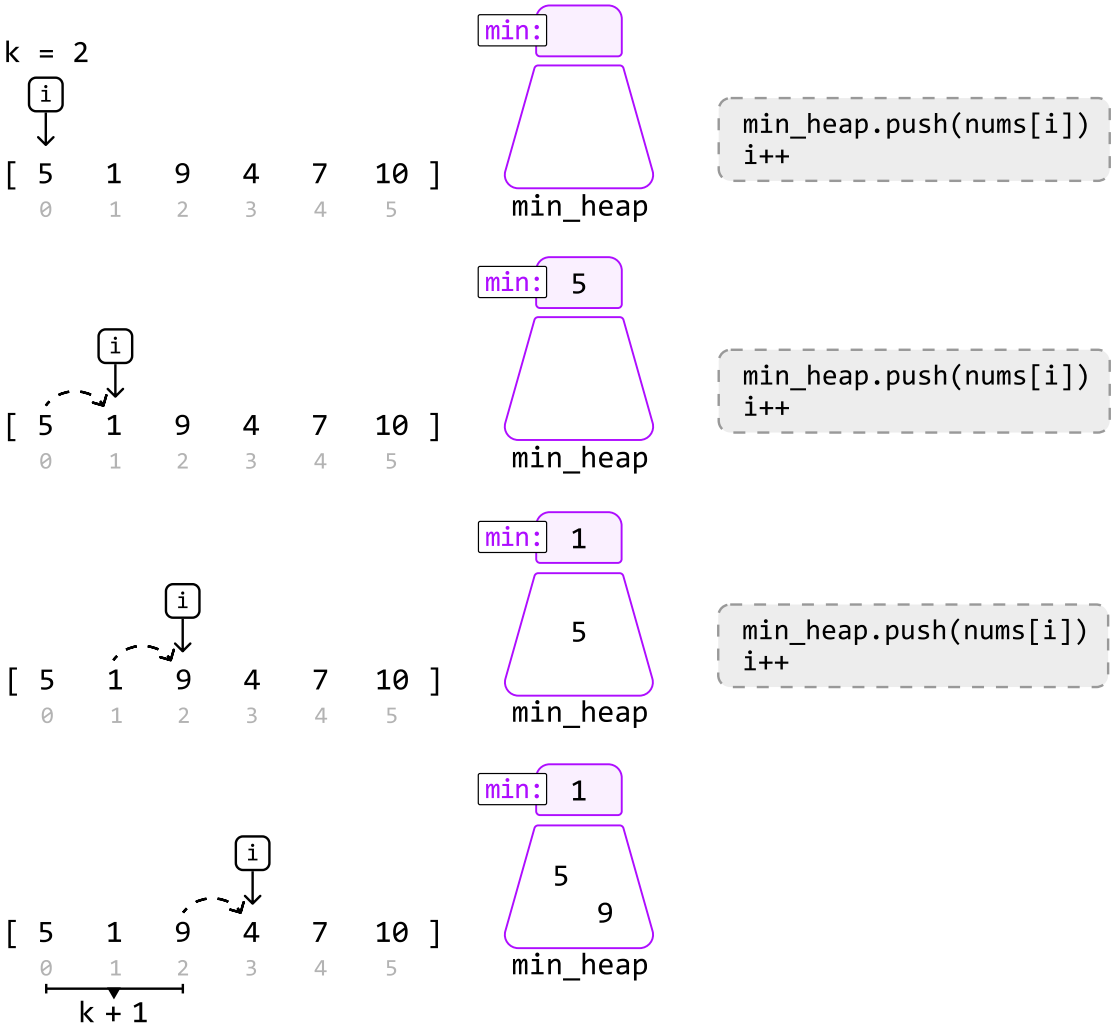
The main inefficiency with this approach is finding the minimum number in the range [i, i + k] at each index i. Linearly searching for it will take $O(k)$ time at each index.

To improve this approach, we'd need a way to efficiently access the minimum value at each of these ranges. A **min-heap** would be perfect for this.

**Min-heap**

For a min-heap to determine the minimum value within each range [i, i + k], it will always need to be populated with the values in these ranges as we iterate through the array. Let's see how this works over the same example.

Before we can determine which value belongs at index 0, we'll need to populate the heap with all the values in the range [0, k], which are the first k + 1 values (where k = 2 in this example):




An alternative way to create a heap of the first `k + 1` elements is to heapify a list of the first `k + 1` elements.

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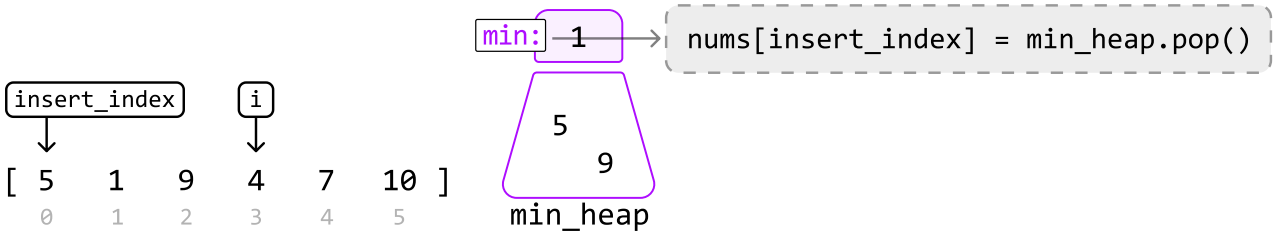
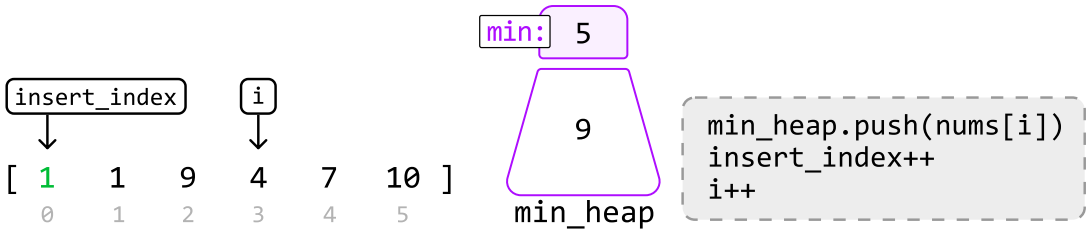
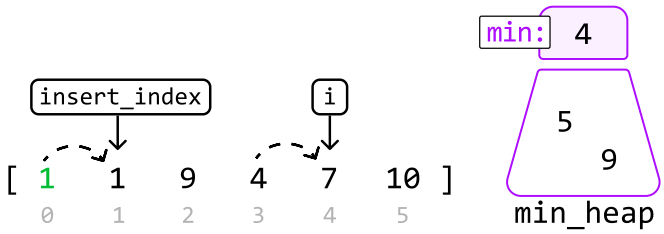
Now, let's begin inserting the smallest elements from the heap into the array, using the `insert_index` pointer. The value that belongs at index 0 in sorted order is the value currently at the top of the heap, which is 1:

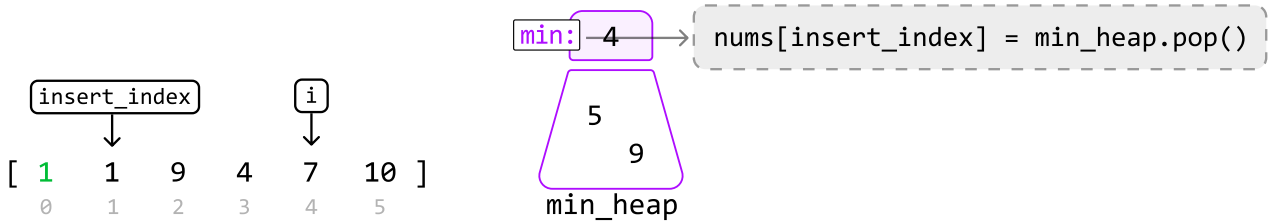


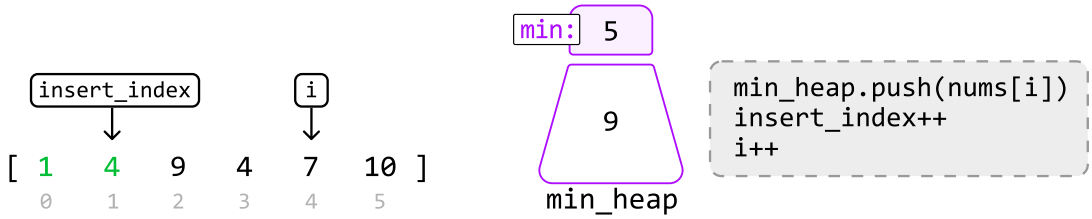

Once we insert 1 at index 0, push the value at index `i` to the heap before incrementing both pointers:

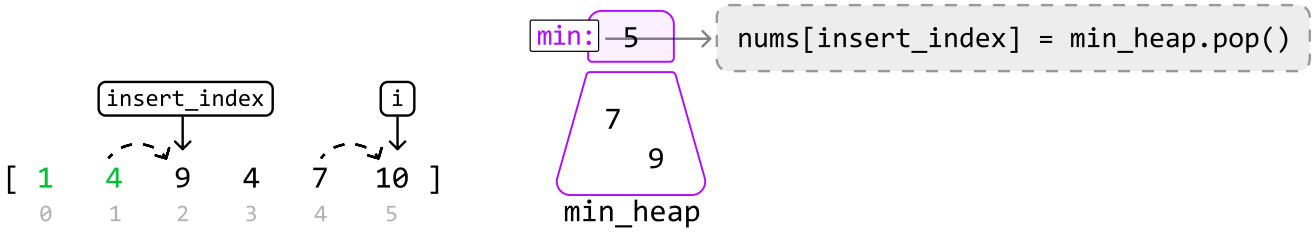


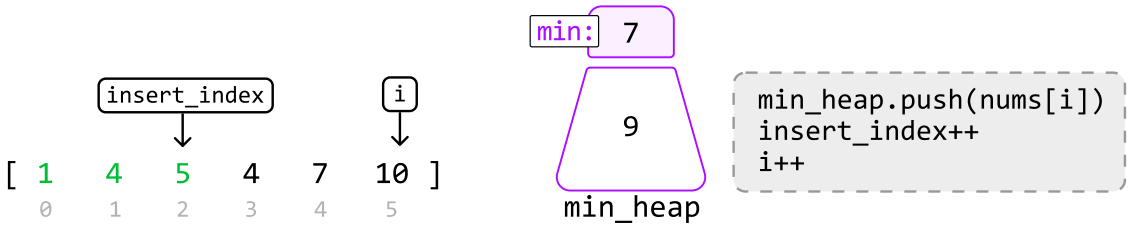



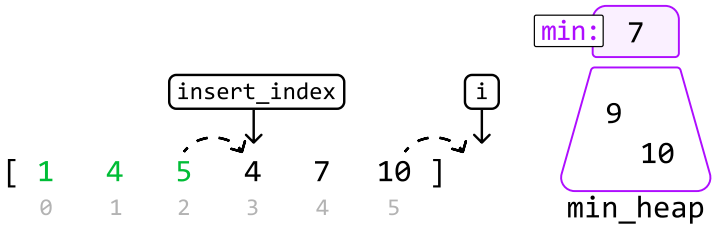

Let's continue this process for the remaining numbers:



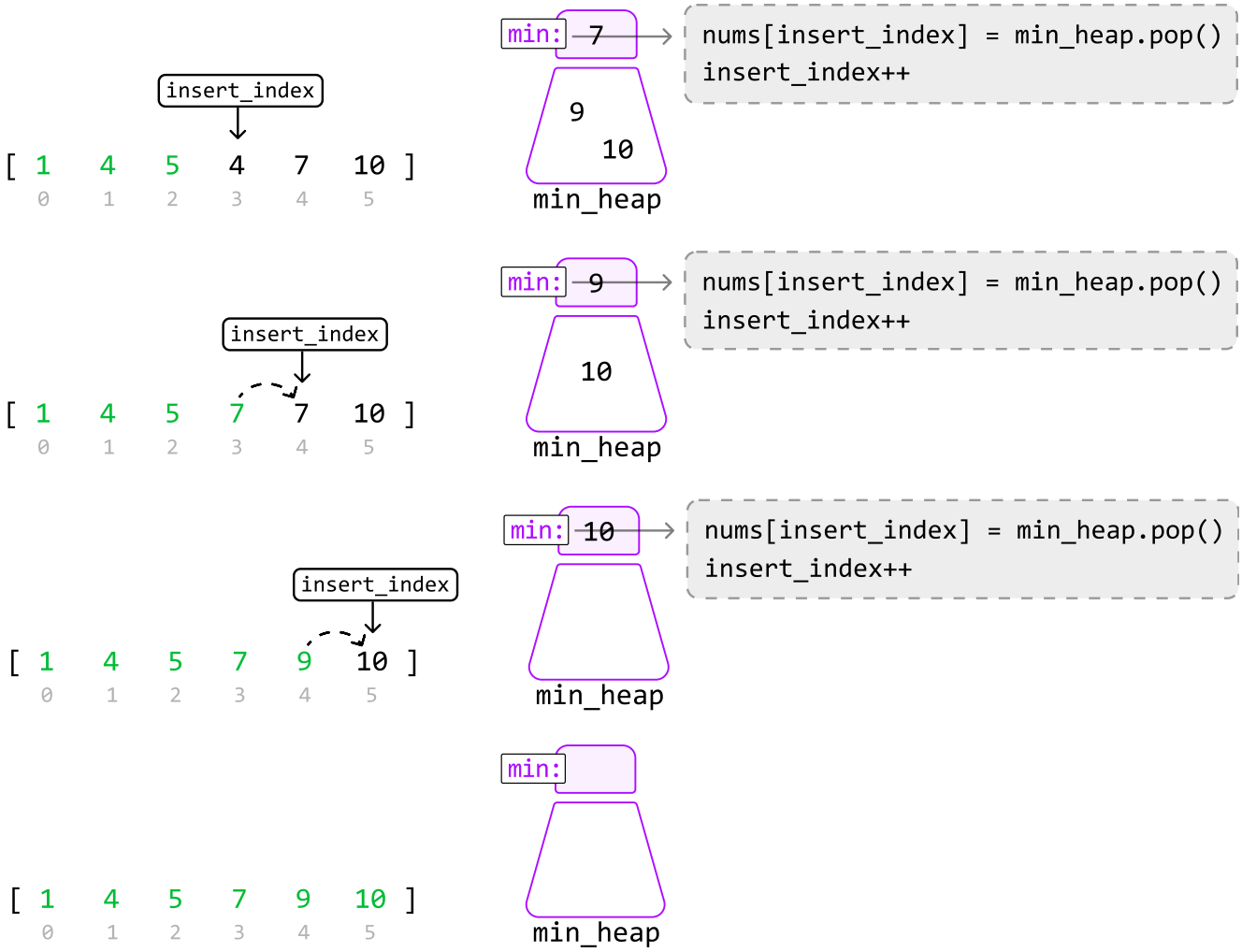



Once there are no more elements to push into the heap, the rest of the array can be sorted by inserting the remaining values from the heap:




Once the heap is empty, the array is sorted.

**Try it yourself**

Write your solution to the problem before checking the reference implementation.

# Implementation

Python JavaScript Java

```
import java.util.ArrayList;
import java.util.PriorityQueue;

class Main {
    public ArrayList<Integer> sort_a_k_sorted_array
        // Populate a min-heap with the first k + 1
        PriorityQueue<Integer> minHeap = new Prior
        int n = nums.size();
```

```
        for (int i = 0; i <= Math.min(k, n - 1); i·
            minHeap.offer(nums.get(i));
        }
        // Replace elements in the array with the r
        // iteration.
        int insertIndex = 0;
        for (int i = k + 1; i < n; i++) {
            nums.set(insertIndex++, minHeap.poll()
            minHeap.offer(nums.get(i));
        }
        // Pop the remaining elements from the heap
        while (!minHeap.isEmpty()) {
            nums.set(insertIndex++, minHeap.poll()
        }
        return nums;
    }
}
```

## Complexity Analysis

**Time complexity:** The time complexity of `sort_a_k_sorted_array` is $O(n \log(k))$, where $n$ denotes the length of the array. Here's why:

- We perform heapify on a min_heap of size $k+1$ which takes $O(k)$ time. Note that $k$ is upper-bounded by $n$ in this operation since the heap won't have more than $n$ values.
- Then, we perform push and pop operations on approximately $n-k$ values using the heap. Since the heap can grow up to a size of $k+1$, each push and pop operations takes $O(\log(k))$ time. Therefore, this loop takes $O(n \log(k))$ time in the worst case.
- The final while-loop runs in $O(k \log(k))$ time since we pop $k+1$ values from the heap. Note that $k$ here is also upper-bounded by $n$.

Therefore, the overall time complexity is $O(k) + O(n \log(k)) + O(k \log(k)) = O(n \log(k))$.

**Space complexity:** The space complexity is $O(k)$ since the heap can grow up to $k+1$ in size.